Why Do We Need an API?

現在遇到的問題

😤 「我電腦可以跑，你不行」

環境不一樣，setup 就花半天

🔒 想用別人的工具，不知道怎麼裝

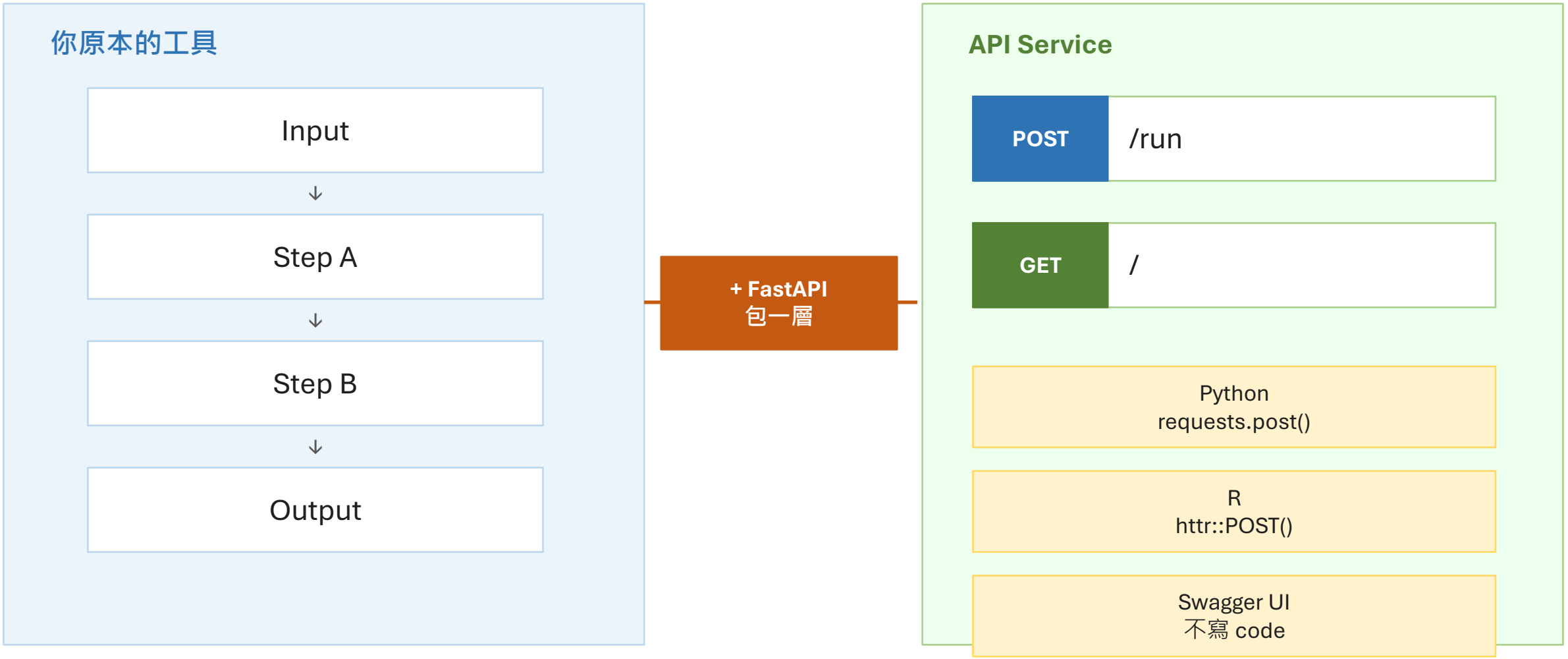
要看 code、要裝套件、要懂對方的環境

🔗 LLM / workflow 很難串別人的東西

沒有統一的呼叫方式，每個工具都不同

這些問題，API 都可以解決

Pipeline as an API Service



不是取代Pipeline，是讓Pipeline 更好用

Why FastAPI?

Python API Framework 比較

	Flask	FastAPI ✅	Django
自動 Swagger 文件	❌	✅	❌
Pydantic input 驗證	❌	✅	❌
學習曲線	低	中	高
適合場景	小工具	**API service**	大型 web app

為什麼是 FastAPI

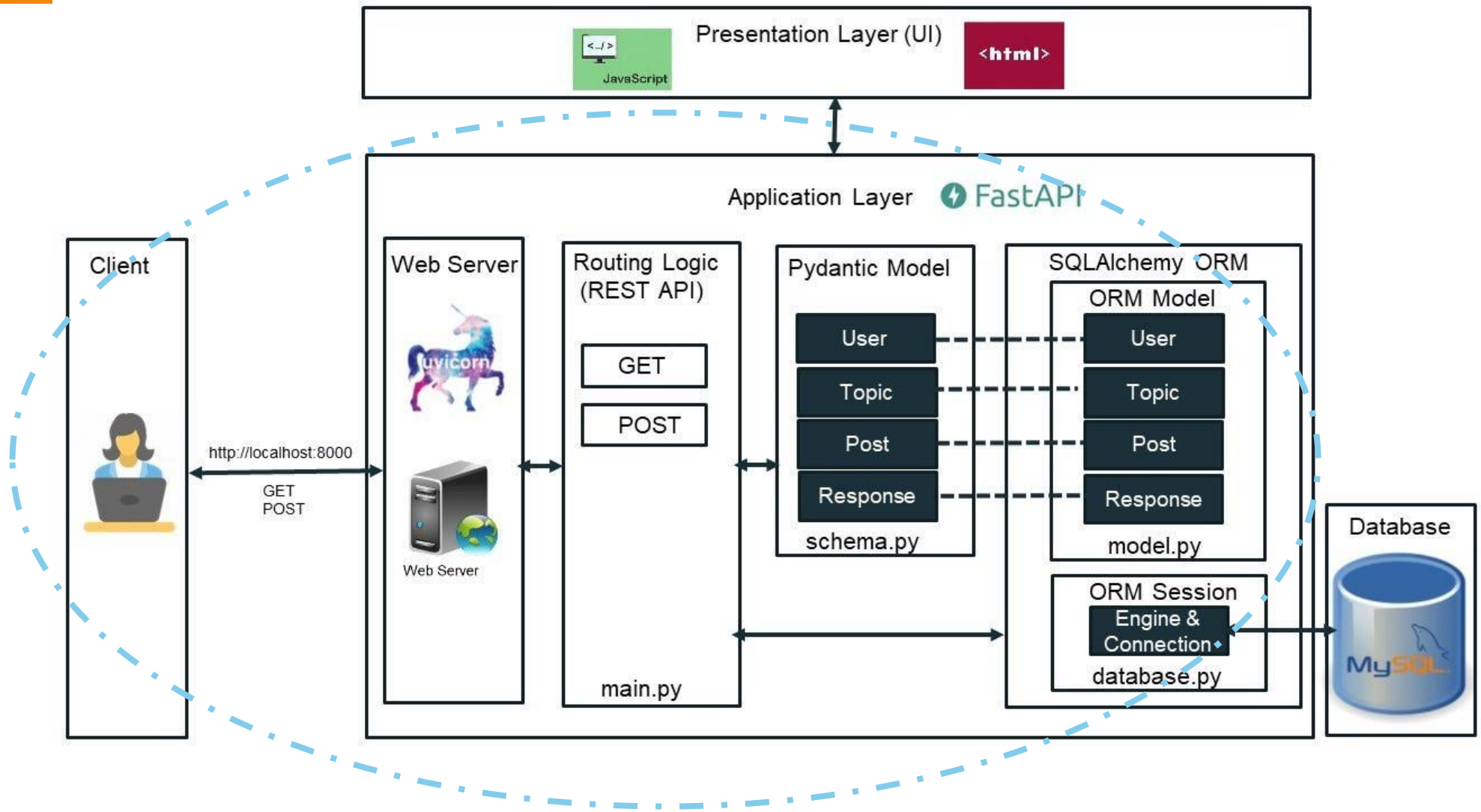
1. 自動產生 Swagger UI — 不寫文件，文件自動出來
2. Pydantic schema — input 驗證免費附贈
3. Async 原生支援 — 同時呼叫多個 API 不卡住

API Intro

Ref. links:

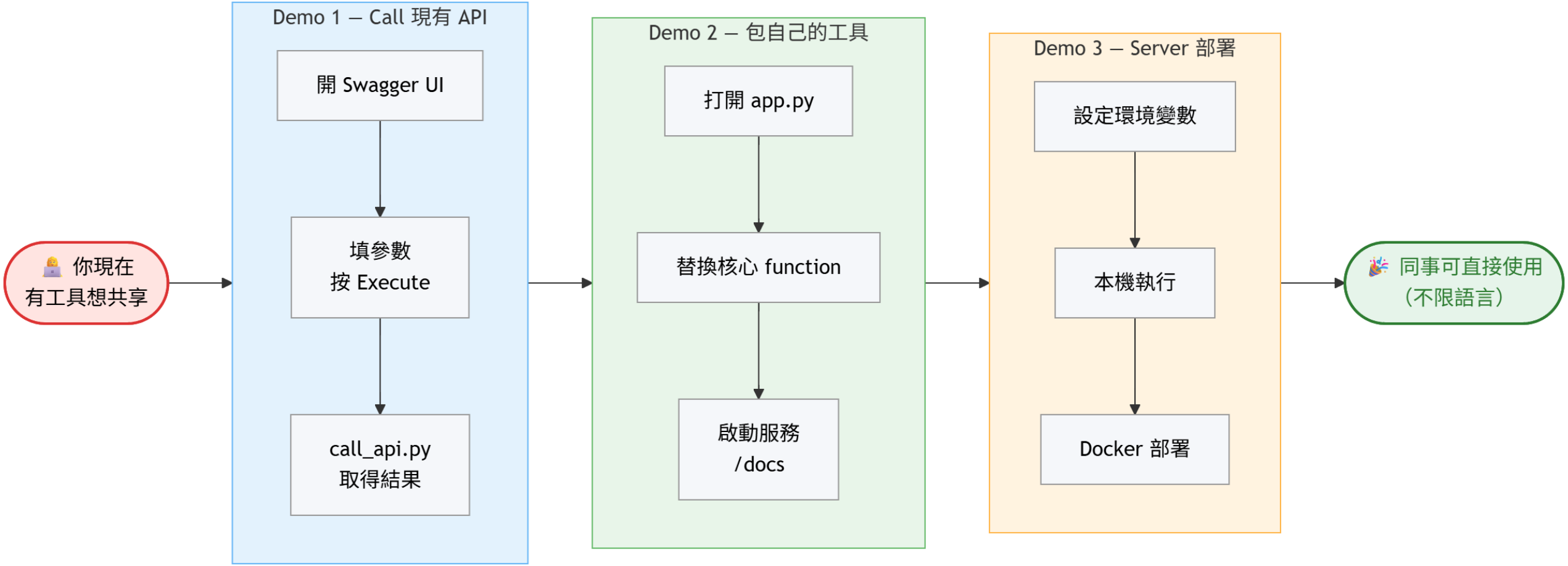
- Python API Tutorial: Getting Started with APIs
- FastAPI – An Intro
- FastAPI Best Practices: A Complete Guide for Building Production-Ready APIs
- FastAPI Best Practices: A Condensed Guide with Examples

- API Intro & System Design



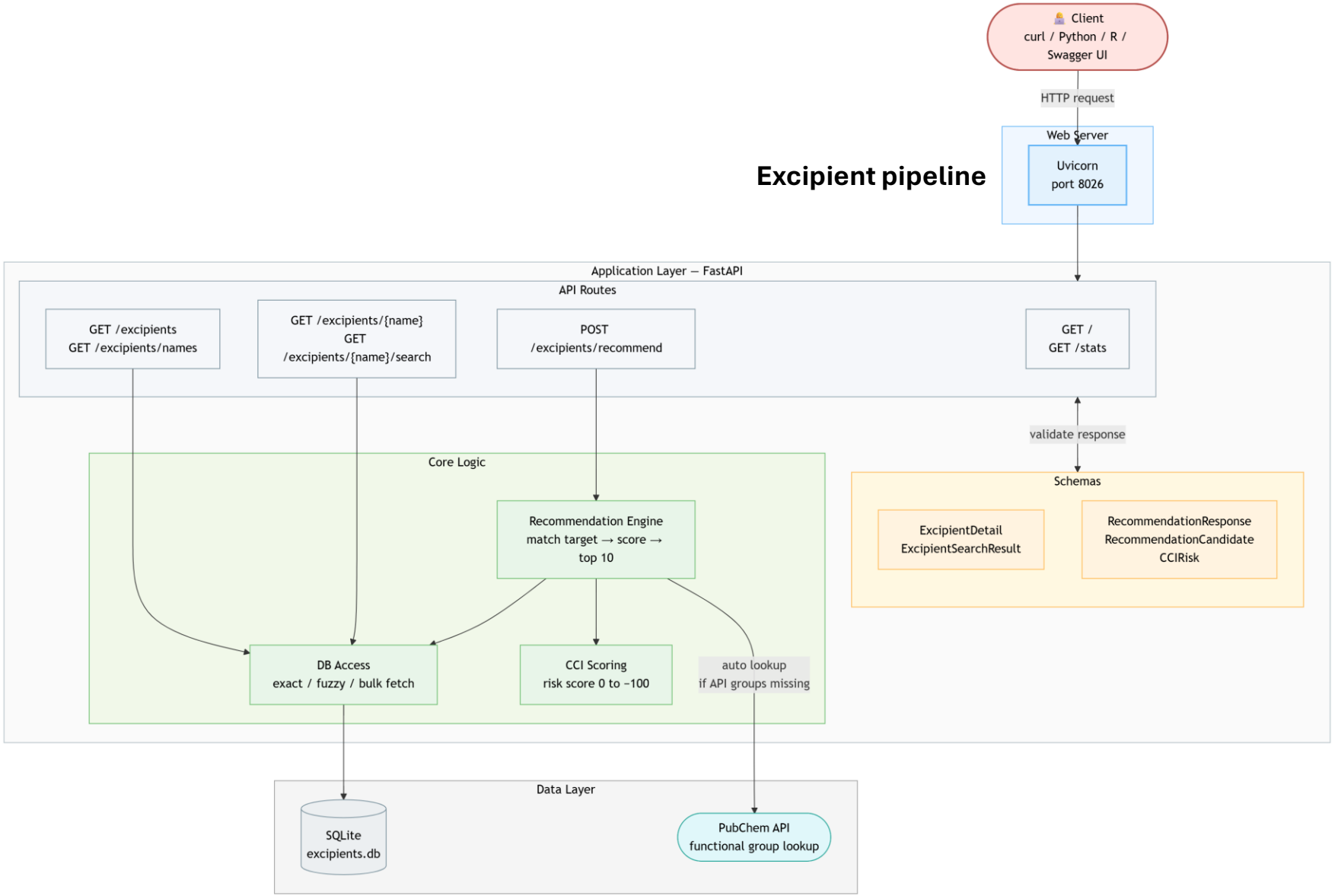
Overview (Tool -> API service)

- In-house tool -> API Service



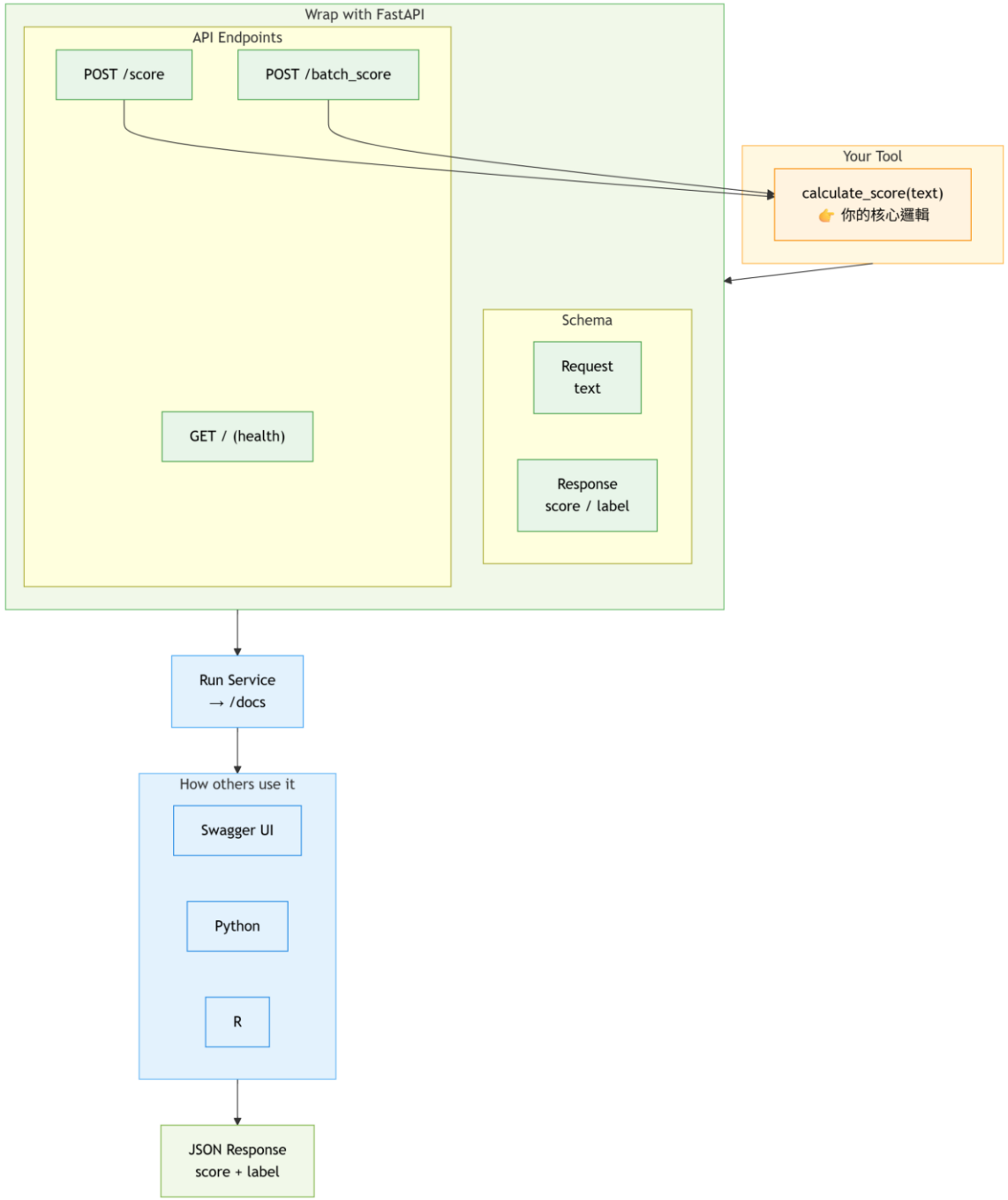
FastAPI Design Example

- Excipient pipeline (Design Flow)



Demo 1 & 2

- FastAPI + Local Deployment (Design Flow)



(Demo 1&2) Demo steps

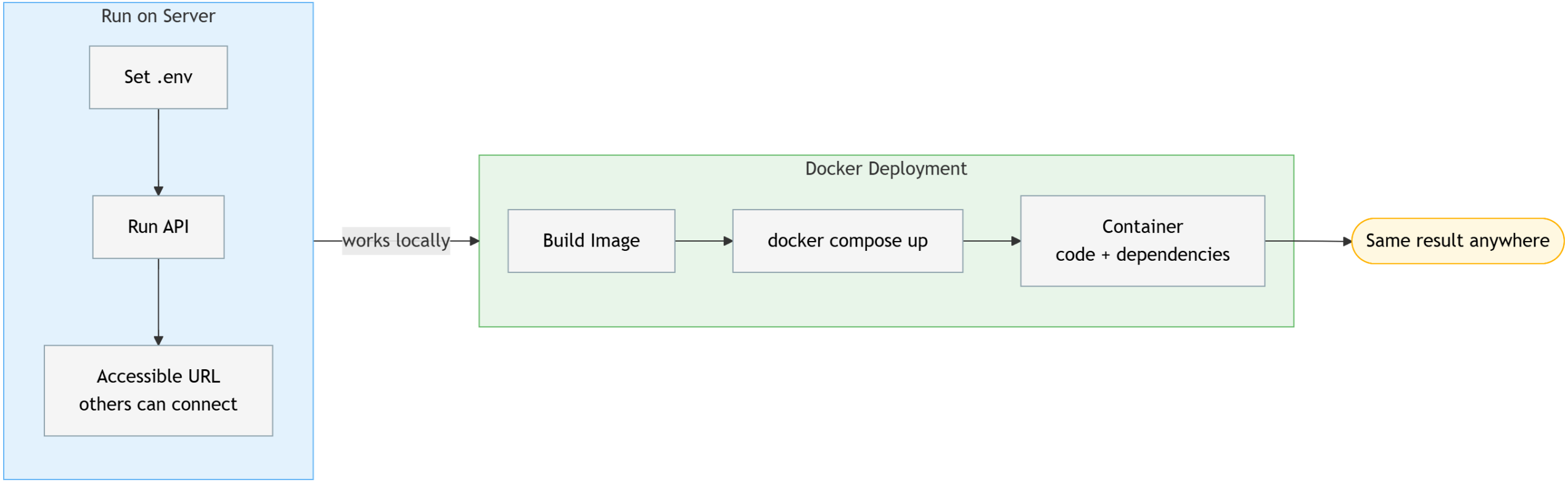
- Demo 1&2 (deploy the demo FastAPI service on my desktop)
  - minimal_service/README.md

```
# example commands
git clone https://github.com/sandyteng0821/starter-pack-fastapi.git

# Demo 2
## environment settings
cd starter-pack-fastapi/
python3 -m venv venv
source venv/bin/activate
pip install -r minimal_service/requirements.txt

## FastAPI service deployment
python3 minimal_service/app.py

# Demo 1 (Call the deployed service)
python3 usage_example/call_api.py
```

Test directory:
/your/file/path/starter-pack-fastapi/

Demo 3

- FastAPI + Docker Deployment (Design Flow)



(Demo 3) Demo steps

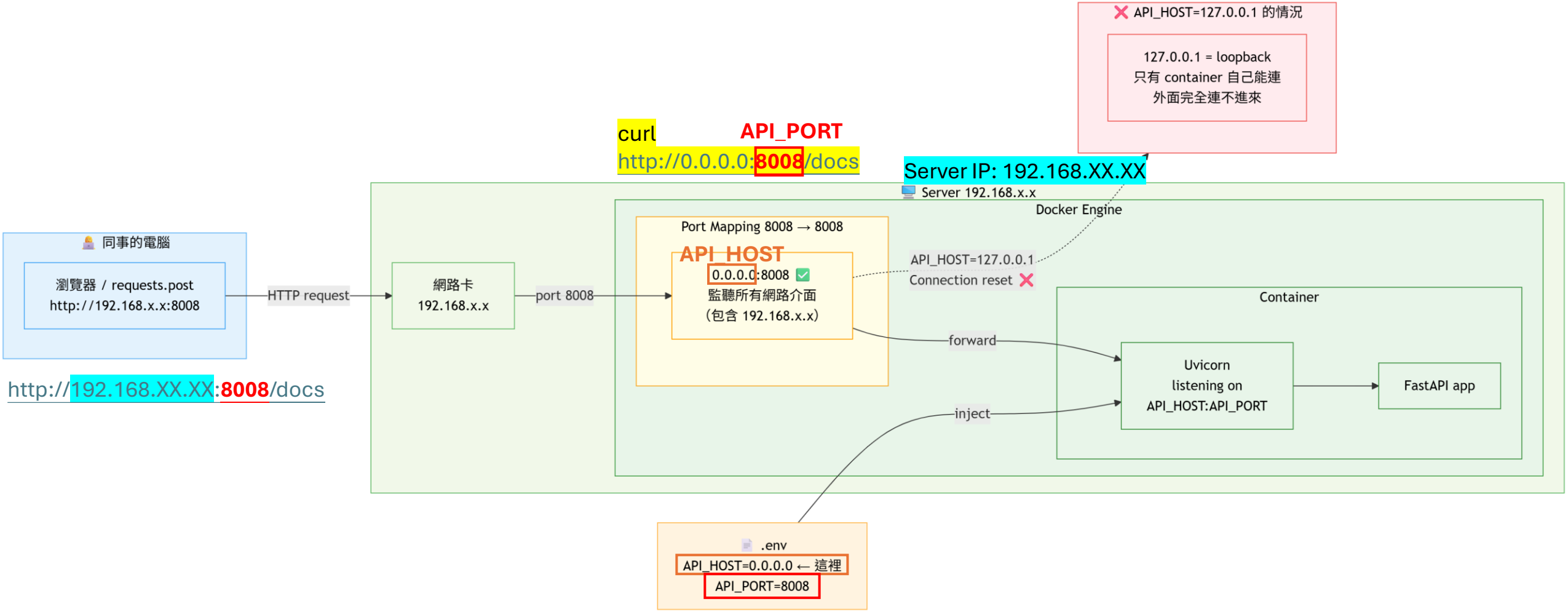
- Demo 3 (deploy the demo FastAPI service on Altos-Workstation-1)
  - advanced_structure/README.md

```
# example commands
git clone https://github.com/sandyteng0821/starter-pack-fastapi.git

# Demo 3
## environment settings
cd starter-pack-fastapi/

## FastAPI service deployment
cd advanced_structure/
cp .env.example .env # 改API_PORT => 8007 -> 8008

# docker deployment
docker compose build --no-cache
docker compose up –d
# deployed service inspection
docker logs -f advanced_structure-api-1
# fastapi service
http://0.0.0.0:8008/docs  # curl http://0.0.0.0:8008
http://192.168.XX.XX:8008/docs
```

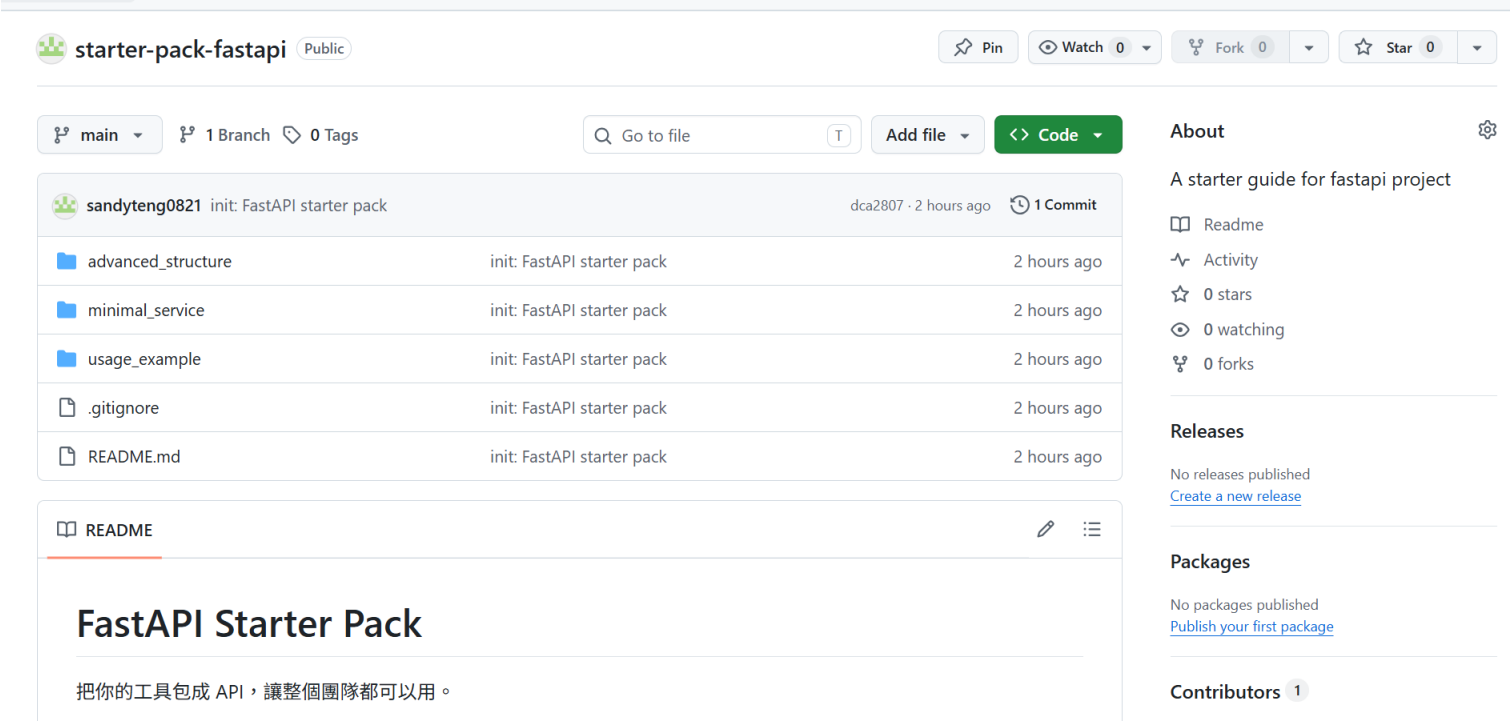
Test directory:
/your/file/path/starter-pack-fastapi/

Demo 3

- Server – Client Design



Shared Links

- Repo
  - https://github.com/sandyteng0821/starter-pack-fastapi.git

```
starter-pack-fastapi/
├── usage_example/      ← 先從這裡開始 (類似 Demo 1, 直接使用現有API)
├── minimal_service/    ← 想自己包的看這個 (Demo 2)
└── advanced_structure/ ← 進階參考 (Demo 3)
```